

N/a

SUMMARY

Goal: build, deploy, test, and use a production-quality personal AI operating system / JARVIS-style agent that can reason, plan, use tools, execute actions, remember context, and report results with strong security, permissions, and observability.

PROJECTS

Personal AI Agent / JARVIS | Long-term product to build a real, reliable, secure personal AI agent for daily use.

- Incremental roadmap: V1 Foundation, V2 Personal productivity, V3 Local computer agent, V4 Autonomous agent.
- V1 capabilities include agent backend, LLM integration, conversation interface, memory, tool system, web research, file handling, GitHub integration, basic browser automation, task management, authentication, permission system, and logging.
- Core architecture includes Planner, Memory, Permissions, Tool Manager, cloud component, and local computer agent.
- Cloud stack preferences: Next.js/React/TypeScript/Tailwind frontend and Python/FastAPI backend with PostgreSQL and Redis.
- Security requirements include authentication, authorization, encrypted secrets, OAuth, rate limiting, approval workflows, secure sessions, input/output validation, and sandboxing where appropriate.
- Observability requirements include tracing requests, plans, tool inputs/outputs, errors, retries, approvals, durations, and token/cost usage.
- Emphasis on controlled tool registry, model-provider abstraction (LLMProvider), and configurable per-tool permissions/risk levels.
- First milestone target: usable dashboard, chat, tasks, memory, web search, file access, GitHub, browser automation, risky-action approvals, and transparent activity reporting.

Railzy | Train-related web/application project referenced as an important current project context for the agent.

- Agent should understand Railzy architecture, repository, deployment, domain, infrastructure, backlog, decisions, bugs, and roadmap.
- Agent workflows include checking pending work and summarizing incomplete tasks by priority.

Explore The Unmapped | YouTube channel workflow to be supported by the agent.

- Support for titles, descriptions, tags, captions, content ideas, social posts, research, publishing preparation, and content calendar.

TECHNICAL SKILLS

AI Agent Architecture

Technical Planning

Full-Stack Engineering

DevOps Engineering

AI Engineering

Product Management

Project Management

Next.js

React

TypeScript

Tailwind CSS

Python

FastAPI
PostgreSQL
pgvector
Redis
Celery
Dramatiq
RQ
Playwright
GitHub Integration
Gmail Integration
Google Calendar Integration
Browser Automation
Task Scheduling
Authentication
Authorization
OAuth
Permission Systems
Audit Logging
Error Handling
Testing
Docker
Nginx
AWS EC2
Linux Servers
Node.js
MongoDB
PM2
VS Code
Windows

DEVELOPMENT ENVIRONMENT

Primary environment and tools: Windows, VS Code, GitHub, Node.js, React, Next.js, TypeScript, Python, FastAPI, MongoDB, PostgreSQL, Docker, PM2, AWS/EC2, Linux servers.

PRODUCT PHILOSOPHY

Useful, reliable, transparent, controllable, extensible, model-independent, and private.

OPERATING PRINCIPLE

Do not build everything at once; define architecture and scope first, then implement incrementally with practical decisions and hands-off continuity.